



UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Third Semester – Spring 2024

Roll No. _____

Time: 3 Hrs.

Marks: 60

Paper: Mathematics A-III

Course Code: MATH-201

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Solve the following:

(6x5=30)

- If $|A| \neq 0$, then find $|A^{-1}|$.
- Find $q(A)$ if $q(x) = x^3 - x^2 + 2x - 3$ and $A = \begin{bmatrix} 1 & 3 \\ -1 & 1 \end{bmatrix}$.
- Find domain and range $T(x, y, z) = (x + y, y + z, x)$.
- Show that T is linear. Also check is it one one? $T(x_1, x_2, x_3) = (x_1 - x_2, x_3)$.
- Let U and W be 2-dimensional subspaces of $\mathbb{R}^3$. Show that $U \cap W \neq \{0\}$.
- Suppose u, v & w are linearly independent vector. Prove that $u + v - 2w, u - v - w, u + w$ are linearly independent.

Q.2. Solve the following:

(5x6=30)

- Check either $T(x, y, z) = (x + y, y + z, x)$ is a matrix transformation also find $R(T)$ and $N(T)$.
- Solve the system by inverting the coefficient matrix.
 $2x_1 - x_2 - x_3 = 4 : 3x_1 - 2x_3 + 4x_2 = 10 : 3x_1 - 2x_2 + 4x_3 = 0$
- If A is $n \times n$ nilpotent matrix. show that $I_n - A$ is nonsingular.
- Find the bases for the subspace of $\mathbb{R}^3$ that is spanned by the vectors:
 $(1, 1, -4), (2, 0, 2)$ & $(2, -1, 3)$
- Find eigenvalue and eigenvectors of the matrix $A = \begin{bmatrix} 1 & 1 \\ -2 & 4 \end{bmatrix}$.

PUACP

Q.1: Short Questions (6×5 = 30)

i) If $|A| \neq 0$, then find $|A^{-1}|$.

Sol: Since A is invertible as $|A| \neq 0$

$$\text{So, } A^{-1}A = I$$

$$|A^{-1}A| = |I|$$

$$|A^{-1}| |A| = 1 \quad (\because |I| = 1) \quad |A^{-1}A| = |A| |A|$$

$$\boxed{|A^{-1}| = \frac{1}{|A|}} \quad \text{As required.}$$

ii) Find $q(A)$ if $q(x) = x^3 - x^2 + 2x - 3$ and $A = \begin{bmatrix} 1 & 3 \\ -1 & 1 \end{bmatrix}$

Sol:

Given $q(x) = x^3 - x^2 + 2x - 3$, $A = \begin{bmatrix} 1 & 3 \\ -1 & 1 \end{bmatrix}$, $q(A) = ?$

First we calculate A^2 and A^3

$$A^2 = A \times A = \begin{bmatrix} 1 & 3 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 1-3 & 3+3 \\ -1-1 & -3+1 \end{bmatrix} = \begin{bmatrix} -2 & 6 \\ -2 & -2 \end{bmatrix}$$

$$A^3 = A^2 \times A = \begin{bmatrix} -2 & 6 \\ -2 & -2 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} -2+6 & -8+6 \\ -2+2 & -6-2 \end{bmatrix} = \begin{bmatrix} -8 & -2 \\ 0 & -8 \end{bmatrix}$$

Now, $q(A) = A^3 - A^2 + 2A - 3I$

$$= \begin{bmatrix} -8 & -2 \\ 0 & -8 \end{bmatrix} - \begin{bmatrix} -2 & 6 \\ -2 & -2 \end{bmatrix} + 2 \begin{bmatrix} 1 & 3 \\ -1 & 1 \end{bmatrix} - 3 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -7 & 0 \\ 0 & -7 \end{bmatrix}$$

iii) Find domain and range $T(x, y, z) = (x+y, y+z, x)$.

Sol: Given $T(x, y, z) = (x+y, y+z, x)$

1- Domain: The domain of a linear transformation is the set of all possible input vectors. Therefore, the domain is $\mathbb{R}^3$.

2- Range: The range of a linear transformation is the set of all possible output vectors.

To find range, expressing linear transformation in terms of a matrix

$$T(1, 0, 0) = (1, 0, 1)$$

$$T(0, 1, 0) = (1, 1, 0)$$

$$T(0, 0, 1) = (0, 0, 0)$$

So, matrix of linear transformation is

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & -1 & 0 \end{bmatrix} \begin{matrix} B_3 \\ R_3 - R_2 \end{matrix}$$

$$\sim \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{matrix} B_3 \\ R_3 + R_2 \end{matrix}$$

So, the vectors are linearly independent and spans $\mathbb{R}^3$.

Thus Range = $\mathbb{R}^3$.

Conclusion: Domain = $\mathbb{R}^3$, Range = $\mathbb{R}^3$.

iv) Show that T is linear. Also check is it one-one?

$$T(x_1, x_2, x_3) = (x_1 - x_2, x_3).$$

(Ex 6.3, Q.6 (ii) M.M Pg-272)

Sol: let $u = (x, y, z)$, $T(u) = (x - y, z)$

$v = (x', y', z')$, $T(v) = (x' - y', z')$

$u + v = (x + x', y + y', z + z')$, $T(u + v) = (x + x' - y - y', z + z')$ — (1)

$T(u) + T(v) = (x + x' - y - y', z + z')$ — (2)

$\rightarrow T(u + v) = T(u) + T(v)$

Now, $ku = (kx, ky, kz)$

$T(ku) = (kx - ky, kz) = k(x - y, z) = kT(u)$

$\rightarrow T(ku) = kT(u)$

Thus, T is linear.

Now, let $u = (x_1, y_1, z_1)$, $v = (x_2, y_2, z_2)$ be any two vectors in $\mathbb{R}^3$.

$T(u) = T(v)$ implies

$(x_1 - y_1, z_1) = (x_2 - y_2, z_2)$

i.e., $x_1 - y_1 = x_2 - y_2$ — (1) and

$z_1 = z_2$

From (1), we cannot infer that $x_1 = x_2$ and $y_1 = y_2$.

Thus, $u \neq v$ as $T(u) = T(v)$ does not imply $u = v$ and

therefore, T is not one-to-one.

v) let U and W be 2-dimensional subspaces of $\mathbb{R}^3$. Show that $U \cap W \neq \{0\}$. (Ex 6.2, Q.17, Pg-253, M.M)

Sol: $\dim U = 2$, $\dim W = 2$

If $U = W$, then $U \cap W \neq \{0\}$ as $\dim U = \dim W = 2$.

Now, suppose $U \neq W$. Then generating sets of U and W

are not the same. Therefore,

$$\dim(U+W) > 2$$

Since U and W are subspaces of $\mathbb{R}^3$, $U+W$ is subspace of $\mathbb{R}^3$.

$$\text{Hence, } \dim(U+W) \leq 3$$

$$\text{Thus } 2 < \dim(U+W) \leq 3$$

$$\text{This implies } \dim(U+W) = 3$$

$$\text{Now, } \dim(U+W) = \dim U + \dim W - \dim(U \cap W)$$

$$\text{i.e., } 3 = 2 + 2 - \dim(U \cap W)$$

Hence, $\dim(U \cap W) = 1$. This means $U \cap W$ contains a nonzero element and so $U \cap W \neq \{0\}$.

vi) Suppose u, v & w are linearly independent vectors. Prove that $u+v-2w, u-v-w, u+w$ are linearly independent. (Ex 6.2 Q.5 ii)

Sol: Suppose a, b, c are scalars and

$$a(u+v-2w) + b(u-v-w) + c(u+w) = 0$$

$$(a+b+c)u + (a-b)v + (-2a-b+c)w = 0$$

Since, u, v and w are linearly independent, we have

$$a+b+c = 0 \quad \text{--- (1)}$$

$$a-b = 0 \quad \text{--- (2)}$$

$$-2a-b+c = 0 \quad \text{--- (3)}$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 1 & -1 & 0 & 0 \\ -2 & -1 & 1 & 0 \end{array} \right] \sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & -2 & -1 & 0 \\ 0 & 1 & 3 & 0 \end{array} \right] \begin{array}{l} \text{By } R_3 + 2R_1 \\ R_2 - R_1 \end{array}$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 0 & 5 & 0 \\ 0 & 1 & 3 & 0 \end{array} \right] \text{ } R_2 + 2R_3$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 1 & 3 & 0 \\ 0 & 0 & 5 & 0 \end{array} \right] \text{ } \text{By } R_{23}$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 1 & 3 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right] \sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right] \text{ Thus, } a=0, b=0, c=0.$$

Hence, given vectors are linearly independent.

Q.2: Long Questions. $15 \times 6 = 30$

i) Check either $T(x, y, z) = (x+y, y+z, x)$ is a matrix transformation.
also find $R(T)$ and $N(T)$.

Let $u = (x_1, y_1, z_1)$, $v = (x_2, y_2, z_2)$, $u+v = (x_1+x_2, y_1+y_2, z_1+z_2)$

$$T(u) = (x_1+y_1, y_1+z_1, x_1), T(v) = (x_2+y_2, y_2+z_2, x_2)$$

$$T(u+v) = (x_1+x_2+y_1+y_2, y_1+y_2+z_1+z_2, x_1+x_2) \quad \text{--- (1)}$$

$$T(u) + T(v) = (x_1+x_2+y_1+y_2, y_1+y_2+z_1+z_2, x_1+x_2) \quad \text{--- (2)}$$

$$\text{From (1) and (2), } T(u+v) = T(u) + T(v) \quad \text{--- (4)}$$

$$\text{Now, } ku = (kx_1, ky_1, kz_1)$$

$$T(ku) = (kx_1+ky_1, ky_1+kz_1, kx_1)$$

$$= k(x_1+y_1, y_1+z_1, x_1) = k(T(u))$$

$$\rightarrow T(ku) = kT(u) \quad \text{--- (3)}$$

From (3) and (4), T is linear transformation.

The standard basis of $\mathbb{R}^3$ consists of the vectors

$$e_1 = (1, 0, 0), e_2 = (0, 1, 0), e_3 = (0, 0, 1)$$

$$\text{Now, } T(e_1) = T(1, 0, 0) = (1, 0, 1) = 1(1, 0, 0) + 0(0, 1, 0) + 1(0, 0, 1)$$

$$T(e_2) = T(0, 1, 0) = (1, 1, 0) = 1(1, 0, 0) + 1(0, 1, 0) + 0(0, 0, 1)$$

$$T(e_3) = T(0, 0, 1) = (0, 1, 0) = 0(1, 0, 0) + 1(0, 1, 0) + 0(0, 0, 1)$$

So, matrix transformation of T is

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

$R(T)$ is the column space of A .

$$\text{So, } R(T) = \{(1, 0, 1), (1, 1, 0), (0, 1, 0)\}$$

$N(T)$ is the null space of A . The null space of A

is set of all vectors $x = (x_1, y_1, z_1)$ such that $Ax = 0$.

$$\text{So, } Ax = 0$$

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \rightarrow \begin{bmatrix} x_1+y_1 \\ y_1+z_1 \\ x_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\rightarrow x_1+y_1=0, y_1+z_1=0, x_1=0$$

$$\text{So, } N(T) = \{(0, 0, 0)\}$$

$$\text{Therefore } R(T) = \{(1, 0, 1), (1, 1, 0), (0, 1, 0)\} \text{ and } N(T) = \{(0, 0, 0)\}$$

ii) Solve the system by inverting the coefficient matrix

$$2x_1 - x_2 - x_3 = 4 : 3x_1 - 2x_2 + 4x_3 = 10 : 3x_1 - 2x_2 + 4x_3 = 0$$

Sol: Consider the system of linear equations

$$\begin{aligned} 2x_1 - x_2 - x_3 &= 4 \\ 3x_1 + 4x_2 - 2x_3 &= 10 \\ 3x_1 - 2x_2 + 4x_3 &= 0 \end{aligned}$$

In matrix form, this system can be written as $AX=B$ where

$$A = \begin{bmatrix} 2 & -1 & -1 \\ 3 & 4 & -2 \\ 3 & -2 & 4 \end{bmatrix}, X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, B = \begin{bmatrix} 4 \\ 10 \\ 0 \end{bmatrix}$$

$$\text{Also, } AX=B \rightarrow X=A^{-1}B$$

Now, we find A^{-1} by elementary row operations

$$\begin{aligned} &\left[\begin{array}{ccc|ccc} 2 & -1 & -1 & 1 & 0 & 0 \\ 3 & 4 & -2 & 0 & 1 & 0 \\ 3 & 2 & 4 & 0 & 0 & 1 \end{array} \right] \sim \left[\begin{array}{ccc|ccc} 1 & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 3 & 4 & -2 & 0 & 1 & 0 \\ 3 & 2 & 4 & 0 & 0 & 1 \end{array} \right] \begin{array}{l} \text{By } R_1 \times \frac{1}{2} \\ R_2 - 3R_1 \\ R_3 - 3R_1 \end{array} \\ &\sim \left[\begin{array}{ccc|ccc} 1 & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 0 & \frac{11}{2} & -\frac{3}{2} & -\frac{3}{2} & 1 & 0 \\ 0 & \frac{7}{2} & \frac{11}{2} & -\frac{3}{2} & 0 & 1 \end{array} \right] \begin{array}{l} R_2 \times \frac{2}{11} \\ R_3 - 7R_2 \end{array} \\ &\sim \left[\begin{array}{ccc|ccc} 1 & 0 & -\frac{6}{11} & \frac{4}{11} & \frac{1}{11} & 0 \\ 0 & 1 & -\frac{1}{11} & -\frac{3}{11} & \frac{2}{11} & 0 \\ 0 & 0 & \frac{128}{11} & -\frac{12}{11} & -\frac{14}{11} & 2 \end{array} \right] \begin{array}{l} R_1 + \frac{6}{11}R_2 \\ R_3 - 128R_2 \end{array} \\ &\sim \left[\begin{array}{ccc|ccc} 1 & 0 & -\frac{6}{11} & \frac{4}{11} & \frac{1}{11} & 0 \\ 0 & 1 & -\frac{1}{11} & -\frac{3}{11} & \frac{2}{11} & 0 \\ 0 & 0 & 1 & \frac{4}{11} & \frac{1}{11} & 2 \end{array} \right] \begin{array}{l} R_1 + \frac{6}{11}R_2 \\ R_3 \times \frac{11}{128} \end{array} \end{aligned}$$

$$\text{As } A^{-1} = \frac{1}{|A|} \text{Adj } A, \text{Adj } A = \begin{bmatrix} A_{11} & A_{21} & A_{31} \\ A_{12} & A_{22} & A_{32} \\ A_{13} & A_{23} & A_{33} \end{bmatrix}$$

$$\begin{aligned} |A| &= \begin{vmatrix} 2 & -1 & -1 \\ 3 & 4 & -2 \\ 3 & -2 & 4 \end{vmatrix} = 2(16-4) + 1(12+6) - 1(-6-12) = 2(12) + 1(18) - 1(-18) \\ &= 24 + 18 + 18 \\ &= 60 \neq 0 \end{aligned}$$

$$A = \begin{bmatrix} 2 & -1 & -1 \\ 3 & 4 & -2 \\ 3 & -2 & 4 \end{bmatrix} \quad A_{11} = (-1)^{1+1} \begin{vmatrix} 4 & -2 \\ -2 & 4 \end{vmatrix} = 1(16-4) = 12$$

$$A_{12} = (-1)^{1+2} \begin{vmatrix} 3 & -2 \\ 3 & 4 \end{vmatrix} = -1(12+6) = -18$$

$$A_{13} = (-1)^{1+3} \begin{vmatrix} 3 & 4 \\ 3 & -2 \end{vmatrix} = 1(-6-12) = -18, \quad A_{21} = (-1)^{2+1} \begin{vmatrix} -1 & -1 \\ -2 & 4 \end{vmatrix} = -(-4-2) = 6$$

$$A_{22} = (-1)^{2+2} \begin{vmatrix} 2 & -1 \\ 3 & 4 \end{vmatrix} = (8+3) = 11, \quad A_{23} = (-1)^{2+3} \begin{vmatrix} 2 & -1 \\ 3 & -2 \end{vmatrix} = -(-4+3) = 1$$

$$A_{31} = (-1)^{3+1} \begin{vmatrix} -1 & -1 \\ 4 & -2 \end{vmatrix} = (2+4) = 6, \quad A_{32} = (-1)^{3+2} \begin{vmatrix} 2 & -1 \\ 3 & -2 \end{vmatrix} = -(-4+3) = 1$$

$$A_{33} = (-1)^{3+3} \begin{vmatrix} 2 & -1 \\ 3 & 4 \end{vmatrix} = (8+3) = 11$$

$$\text{Adj } A = \begin{bmatrix} 12 & 6 & 6 \\ -18 & 11 & 1 \\ -18 & 1 & 11 \end{bmatrix}$$

Thus, $A^{-1} = \frac{1}{|A|} \text{Adj } A = \frac{1}{60} \begin{bmatrix} 12 & 6 & 6 \\ -18 & 11 & 1 \\ -18 & 1 & 11 \end{bmatrix}$

$$\text{Now, } X = A^{-1}B = \frac{1}{60} \begin{bmatrix} 12 & 6 & 6 \\ -18 & 11 & 1 \\ -18 & 1 & 11 \end{bmatrix} \begin{bmatrix} 4 \\ 10 \\ 0 \end{bmatrix} = \frac{1}{60} \begin{bmatrix} 48+60+0 \\ -72+110+0 \\ -72+10+0 \end{bmatrix} = \frac{1}{60} \begin{bmatrix} 108 \\ 38 \\ -62 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 108/60 \\ 38/60 \\ -62/60 \end{bmatrix} = \begin{bmatrix} 9/5 \\ 19/30 \\ -31/30 \end{bmatrix}$$

$$\boxed{x_1 = \frac{9}{5}, x_2 = \frac{19}{30}, x_3 = -\frac{31}{30}}$$

iii) If A is $n \times n$ nilpotent matrix, show that $I_n - A$ is non-singular.

Sol: (Linear Algebra Example 5 ch 1 Pg-48 2R BHATTI)

Let A be a nilpotent matrix of index p , then $A^p = 0$.

Consider

$$\begin{aligned} (I_n - A)(I_n - A)^{-1} &= (I_n - A)(I_n + A + A^2 + \dots + A^{p-1} + \dots) \quad (\text{Using Binomial Theorem}) \\ &= (I_n - A)(I_n + A + A^2 + \dots + A^{p-1}) \\ &= I_n(I_n + A + A^2 + \dots + A^{p-1}) - A(I_n + A + A^2 + \dots + A^{p-1}) \\ &= (I_n + A + A^2 + \dots + A^{p-1}) - (AI_n + A^2 + A^3 + \dots + A^p) \end{aligned}$$

$$= (I_n + A + A^2 + \dots + A^{p-1}) - (A + A^2 + A^3 + \dots + A^{p-1}) \quad \because A^p = A, A^p = 0.$$

$$= I_n$$

Again

$$(I_n - A)^{-1}(I_n - A) = (I_n + A + A^2 + \dots + A^{p-1})(I_n - A) \quad \because A^p = 0$$

$$= (I_n + A + A^2 + \dots + A^{p-1})(I_n - A)$$

$$= (I_n + A + A^2 + \dots + A^{p-1})I_n - (I_n + A + A^2 + \dots + A^{p-1})A$$

$$= (I_n + A + A^2 + \dots + A^{p-1}) - (I_n A + A^2 + A^3 + \dots + A^p)$$

$$= (I_n + A + A^2 + \dots + A^{p-1}) - (A + A^2 + A^3 + \dots + A^{p-1}) \quad \because I_n A = A, A^p = 0.$$

$$= I_n$$

Hence, $(I_n - A)(I_n - A)^{-1} = (I_n - A)^{-1}(I_n - A) = I_n$

Thus, $(I_n - A)^{-1}$ is the inverse of $(I_n - A)$ and so, $(I_n - A)$ is non-singular.

iv) Find the basis for the subspace of $\mathbb{R}^3$ that is spanned by the vectors: $(1, 1, -4)$, $(2, 0, 2)$ & $(2, -1, 3)$. (Like Example 9 Pg-230 L.A. Bhatti.)

Sol: Let $a, b, c \in \mathbb{R}$ and consider

$$a(1, 1, -4) + b(2, 0, 2) + c(2, -1, 3) = 0$$

$$(a, a, -4a) + (2b, 0, 2b) + (2c, -c, 3c) = 0$$

$$(a + 2b + 2c, a - c, -4a + 2b + 3c) = 0$$

$$\rightarrow a + 2b + 2c = 0$$

$$a - c = 0$$

$$-4a + 2b + 3c = 0$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 2 & 0 \\ 1 & 0 & -1 & 0 \\ -4 & 2 & 3 & 0 \end{array} \right] \sim \left[\begin{array}{ccc|c} 1 & 2 & 2 & 0 \\ 0 & -2 & -3 & 0 \\ 0 & 10 & 11 & 0 \end{array} \right] \begin{array}{l} \text{By } A_2 - R_1 \\ R_3 + 4R_1 \end{array}$$

$$\sim \left[\begin{array}{ccc|c} 1 & 2 & 2 & 0 \\ 0 & 1 & -\frac{3}{2} & 0 \\ 0 & 10 & 11 & 0 \end{array} \right] \text{By } \frac{R_2}{-2}$$

$$\sim \left[\begin{array}{ccc|c} 1 & 2 & 2 & 0 \\ 0 & 1 & -\frac{3}{2} & 0 \\ 0 & 0 & 26 & 0 \end{array} \right] \text{By } R_3 - 10R_2$$

$$\sim \left[\begin{array}{ccc|c} 1 & 2 & 2 & 0 \\ 0 & 1 & -\frac{3}{2} & 0 \\ 0 & 0 & 26 & 0 \end{array} \right] \sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right]$$

$\rightarrow a = 0, b = 0, c = 0$, Thus, given vectors are linearly independent and form basis for $\mathbb{R}^3$.

V) Find eigenvalue and eigenvectors of the matrix $A = \begin{bmatrix} 1 & 1 \\ -2 & 4 \end{bmatrix}$.
 Sol: (Example 15 ch 7 Pg. 301 Mathematical Methods).

The characteristic equation of A is

$$p(\lambda) = |A - \lambda I| = \begin{vmatrix} 1-\lambda & 1 \\ -2 & 4-\lambda \end{vmatrix}$$

$$= (1-\lambda)(4-\lambda) + 2 = 0$$

$$\rightarrow \lambda^2 - 5\lambda + 6 = 0 \quad \text{--- (1)}$$

$$\lambda^2 - 3\lambda - 2\lambda + 6 = 0$$

$$\lambda(\lambda-3) - 2(\lambda-3) = 0$$

$$(\lambda-3)(\lambda-2) = 0$$

The roots of (1) are

$\lambda_1 = 2$, $\lambda_2 = 3$. which are eigenvalues of A.

The eigenvector of A corresponding to eigenvalue $\lambda_1 = 2$ is

$$A v_1 = 2 v_1$$

$$\begin{bmatrix} 1 & 1 \\ -2 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = 2 \begin{bmatrix} x_1 \\ y_1 \end{bmatrix}$$

$$\text{where } v_1 = \begin{bmatrix} x_1 \\ y_1 \end{bmatrix}$$

$$\text{Thus, } x_1 + y_1 = 2x_1 \quad \text{--- (2)}$$

$$-2x_1 + 4y_1 = 2y_1 \quad \text{--- (3)}$$

From (2), $x_1 = y_1$ and (3)

gives $y_1 = x_1$

$$\text{Then, } v = \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_1 \end{bmatrix} = y_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Then, $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ is an eigenvector.

All other eigenvectors are scalar multiples of v_1 .

The eigenspace E_2 of A corresponding to eigenvalue $\lambda_1 = 2$ is generated by $v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. It is

subspace of R_2 .

The eigenvector v_2 of A, corresponding to the eigenvalue $\lambda_2 = 3$, is given by

$$A v_2 = 3 v_2$$

That is

$$\begin{bmatrix} 1 & 1 \\ -2 & 4 \end{bmatrix} \begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = 3 \begin{bmatrix} x_2 \\ y_2 \end{bmatrix}$$

$$\text{So, } \begin{cases} x_2 + y_2 = 3x_2 \\ -2x_2 + 4y_2 = 3y_2 \end{cases} \rightarrow y_2 = 2x_2$$

For $x_1 = 1$, $y_2 = 2$.

$$\text{So, } v_2 = \begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = \begin{bmatrix} x_2 \\ 2x_2 \end{bmatrix} = x_2 \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$v_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ is eigenvector for $\lambda_2 = 3$.

Other eigenvectors corresponding to

$\lambda_2 = 3$ are scalar multiples of $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$.

The eigenspace E_3 corresponding to $\lambda_2 = 3$ is generated by $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$.